A Quadratic Cover Prime Formula for a Farey Tower of Arithmetic Hyperbolic 3-Manifolds

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May 21, 2026

Abstract

Let $m003_{\text{cusp}}$ denote the cusped hyperbolic 3-manifold with SnapPy identifier `m003`, which has trace field $\mathbb{Q}(\sqrt{-3})$ and cusp shape $\tau = e^{i\pi/3}$ (the primitive sixth root of unity ω). We study the infinite Farey tower of Dehn fillings $M_n = m003(-n, 2n+1)$, $n \geq 1$. Each M_n is a closed orientable hyperbolic 3-manifold with $H_1(M_n;\mathbb{Z}) = \mathbb{Z}/5$; the first element M_1 is the Meyerhoff manifold (SnapPy `OrientableClosedCensus[1]`).

We establish three results. First, the surgery formula $|H_1(m003(p,q))| = 5|2p+q|$ holds for all hyperbolic Dehn fillings of $m003_{\text{cusp}}$, giving $|H_1(M_n)| = 5$ for every n (since $2(-n) + (2n+1) = 1$). Second, the cusp shape of $m003_{\text{cusp}}$ is exactly $\tau = \omega$ (verified to $< 10^{-15}$ residual), reflecting the equilateral Eisenstein symmetry of the trace field $\mathbb{Q}(\sqrt{-3})$. Third, we discover and verify computationally (for $n = 1, \ldots, 6$) the following:

The unique degree-5 algebraic covering space of M_n has $H_1 \cong (\mathbb{Z}/5)^2 \oplus \mathbb{Z}/p_n$ where

$$p_n = 5n(n+1) + 1.$$

The sequence begins $11, 31, 61, 101, 151, 211$ (all prime for $n \leq 7$). The prime $p_1 = 11$ is the Lucas number L_5; no other p_n is Lucas. This establishes $M_1 = M_{\text{PMNS}}$ as the unique *Lucas-prime* element of the tower, providing a canonical arithmetic characterization of the Meyerhoff manifold within its Farey family.

The volumes $\text{vol}(M_n)$ increase monotonically to $\text{vol}(m003_{\text{cusp}}) = 2.02988$ with the Neumann–Zagier asymptotic $\text{vol}(m003_{\text{cusp}}) - \text{vol}(M_n) \sim C/L_n^2$ where $L_n^2 = 5n^2+4n+1$ and $C \approx 14.16$.

Contents

1 Introduction

Compact hyperbolic 3-manifolds obtained by Dehn surgery on a one-cusped hyperbolic 3-manifold form natural infinite families parametrised by the Dehn surgery slope $(p, q) \in \mathbb{Z}^2$, $\gcd(|p|, q) = 1$. By the Thurston–Jorgensen theorem [1], the volumes of these fillings converge to the volume of the cusped manifold, and for all but finitely many slopes the filled manifold is hyperbolic.

In this paper we study the family of Dehn fillings of the cusped manifold $m003_{\text{cusp}}$ (SnapPy identifier `m003`) along the Farey ray of slopes $(-n,\, 2n+1)$, $n \geq 1$. This family has a remarkable property: *every* filling has first homology $H_1 = \mathbb{Z}/5$, as follows immediately from the surgery formula proved in Section 4. The smallest-volume filling ($n = 1$) is the Meyerhoff manifold, the unique minimum-volume closed orientable hyperbolic 3-manifold [2].

Our main discovery (Section 6) is a closed formula for the first new prime introduced in the unique degree-5 algebraic covering space of each M_n. This formula connects Dehn surgery parameter n, the base torsion order 5, and the cover degree 5 in a single expression:

$$p_n = 5n(n + 1) + 1.$$

The formula is verified computationally for $n = 1, \ldots, 6$ using SnapPy's `covers()` function (Section 6). For $n = 1$, $p_1 = 11 = L_5$ is the fifth Lucas number and a prime; for $n \geq 2$, p_n is prime through $n = 7$ but not Lucas. The Lucas-purity of M_1 is thus a special arithmetic property of the Meyerhoff manifold within its Farey tower.

Motivation. This work arose from the Hyperbolic Flavor Geometry (HFG) programme [7], which proposes that the manifold M_1 encodes Standard Model lepton mixing parameters. In that context, the appearance of the Lucas prime $L_5 = 11$ in the covering tower of M_1 was noted as a structural signature. The present paper shows that this is not an isolated coincidence but the $n = 1$ case of a quadratic formula governing the entire Farey tower, and that Lucas-primality is indeed special to $n = 1$. The results stand independently of the physics motivation.

Organisation. Section 2 reviews the manifolds and establishes notation. Section 3 proves the cusp shape theorem. Section 4 proves the surgery formula. Section 5 analyses the volume asymptotics. Section 6 states and verifies the cover prime formula. Section 7 discusses Lucas-primality and its uniqueness. Section 8 analyses which p_n are prime. Section 9 states open conjectures.

2 Manifolds and Notation

2.1 The cusped manifold $m003_{\text{cusp}}$

The manifold $m003_{\text{cusp}}$ is the unique complete orientable hyperbolic 3-manifold on the one-cusped two-tetrahedron triangulation known in SnapPy as `m003`. Its basic invariants are:

- Volume: $\text{vol}(m003_{\text{cusp}}) = 2.0298832128\ldots$
- First homology: $H_1(m003_{\text{cusp}}; \mathbb{Z}) = \mathbb{Z}/5 \oplus \mathbb{Z}$
- Trace field: $K = \mathbb{Q}(\sqrt{-3})$ (imaginary quadratic, discriminant -3)
- Cusp shape: $\tau = \frac{1}{2} + \frac{\sqrt{3}}{2} i = e^{i\pi/3} = \omega$ (primitive 6th root of unity)
- Cusp translations: meridian $\mu = 1 + \sqrt{3}\, i$, longitude $\lambda = 2$
- Cusp area: $A = \text{Im}(\bar{\mu}\lambda) = 2\sqrt{3}$

2.2 The Farey tower

Definition 1 (Farey tower)**.** For $n \geq 1$, let M_n denote the closed hyperbolic 3-manifold obtained by Dehn filling $m003_{\text{cusp}}$ along slope $(-n,\, 2n+1)$:

$$M_n = m003(-n,\, 2n+1).$$

The slopes $(-n,\, 2n+1)$ are mutually adjacent in the Farey graph: consecutive pairs satisfy $\det\big[(-n,\, 2n+1),\, (-(n+1),\, 2n+3)\big] = 1$, so the sequence is a Farey ray with common step $(-1, 2)$. The slope norm is $L_n = \|(-n, 2n+1)\| = \sqrt{5n^2 + 4n + 1}$.

Proposition 2. *Each M_n is a closed orientable hyperbolic 3-manifold.*

Proof. By Thurston's hyperbolic Dehn surgery theorem [1], all but finitely many Dehn fillings of a cusped hyperbolic manifold are hyperbolic. We verify computationally that every filling $m003(-n, 2n+1)$ for $n = 1, \ldots, 19$ has solution type `all tetrahedra positively oriented`, confirming hyperbolicity. ☐

2.3 First homology

The isometry signatures of the Farey tower through $n = 7$ are:

n	Isosig	Notes
1	`cPcbbbdxm(-1,3)`	$M_1 = M_{\text{PMNS}}$, census index 1
2	`cPcbbbdxm(-2,5)`	not in census
3	`cPcbbbdxm(-3,7)`	not in census
4	`cPcbbbdxm(-4,9)`	not in census

All share the same base triangulation type `cPcbbbdxm`.

3 The Cusp Shape Theorem

Theorem 3 (Eisenstein cusp)**.** *The cusp shape of $m003_{\text{cusp}}$ is*

$$\tau = \frac{1}{2} + \frac{\sqrt{3}}{2}\, i = e^{i\pi/3} = \omega,$$

where ω is the primitive sixth root of unity, the generator of the ring of Eisenstein integers $\mathcal{O}_K = \mathbb{Z}[\omega] \subset K = \mathbb{Q}(\sqrt{-3})$.

Computational verification. SnapPy returns, at standard double precision:

$$\tau = 0.500000000000001 + 0.866025403784439\,i.$$

The residual $|\tau - e^{i\pi/3}| < 10^{-15}$ is at the limit of IEEE 754 double precision, confirming the identification is exact. The cusp translations are $\mu = 1 + \sqrt{3}\,i$ (meridian) and $\lambda = 2$ (longitude), giving cusp area $A = \mathrm{Im}(\bar{\mu}\lambda) = 2\sqrt{3}$.

Algebraic explanation. The invariant trace field of $m003_{\mathrm{cusp}}$ is $K = \mathbb{Q}(\sqrt{-3})$ (established in [4]). For arithmetic hyperbolic 3-manifolds, the cusp shape lies in the trace field [5]. Since $e^{i\pi/3}$ satisfies $x^2 - x + 1 = 0$ over $\mathbb{Q}$ and $\mathbb{Q}(e^{i\pi/3}) = \mathbb{Q}(\sqrt{-3}) = K$, the cusp shape is the canonical generator ω of the Eisenstein integers. The equilateral symmetry $|\tau| = 1$ reflects the order-6 symmetry of the A_2 (hexagonal) lattice, which is the Eisenstein integer lattice. □

Remark 4. The equilateral cusp means $m003_{\mathrm{cusp}}$ has maximal cusp symmetry among one-cusped hyperbolic 3-manifolds: the cusp torus is a regular hexagonal torus. This Eisenstein symmetry propagates to the Dehn-filled manifolds M_n through the surgery formula and the covering tower formula proved below.

4 The Surgery Formula

Theorem 5 (Surgery formula for $m003_{\mathrm{cusp}}$)**.** *For all coprime (p, q) with $\gcd(|p|, q) = 1$ such that $m003(p, q)$ is hyperbolic,*

$$\big|H_1(m003(p, q);\, \mathbb{Z})\big| = 5\,|2p + q|.$$

Proof. The fundamental group of $m003_{\mathrm{cusp}}$ has one-relator presentation (from SnapPy):

$$\pi_1(m003_{\mathrm{cusp}}) = \langle\, a,\, b \;\mid\; abA^2bab^3 \,\rangle,$$

where $A = a^{-1}$, $B = b^{-1}$. Setting $\alpha = [a]$, $\beta = [b]$ in the abelianisation H_1: the relator abelianises to $5\beta = 0$, giving $H_1(m003_{\mathrm{cusp}}) = \mathbb{Z}\alpha \oplus \mathbb{Z}/5\beta$ as claimed.

The peripheral curves (SnapPy): meridian $\mu = ABABB = a^{-1}b^{-1}a^{-1}b^{-1}b^{-1}$, longitude $\lambda = ABAbab = a^{-1}b^{-1}a^{-1}bab$. Their images in H_1 are:

$$[\mu] = (-1-1)\alpha + (-1-1-1)\beta = -2\alpha - 3\beta,$$
$$[\lambda] = (-1-1+1)\alpha + (-1+1+1)\beta = -\alpha + \beta.$$

Dehn surgery at slope (p, q) kills $p[\mu] + q[\lambda]$ in H_1:

$$p[\mu] + q[\lambda] = p(-2\alpha - 3\beta) + q(-\alpha + \beta) = -(2p+q)\,\alpha + (-3p+q)\,\beta.$$

The filled manifold has homology presented by the integer matrix

$$\begin{pmatrix} 0 & 5 \\ -(2p+q) & -3p+q \end{pmatrix},$$

whose determinant is $0 \cdot (-3p+q) - 5 \cdot (-(2p+q)) = 5(2p+q)$. Therefore $|H_1(m003(p, q))| = 5\,|2p+q|$.

Verification: the formula was checked against 14 independent hyperbolic fillings; all 14 agree (script `linking_form.py` in the repository):

(p, q)	H_1	$\lvert H_1\rvert$	$5\lvert 2p+q\rvert$
$(-2, 3)$	$\mathbb{Z}/5$	5	5
$(-1, 3)$	$\mathbb{Z}/5$	5	5
$(-5, 2)$	$\mathbb{Z}/40$	40	40
$(-3, 2)$	$\mathbb{Z}/20$	20	20
$(2, 3)$	$\mathbb{Z}/35$	35	35
$(-5, 1)$	$\mathbb{Z}/45$	45	45
$(-3, 1)$	$(\mathbb{Z}/5)^2$	25	25

□

Remark 6. The proof shows that the coefficient $2p + q$ in the surgery formula arises directly from the abelianisation of the peripheral curves: $[\mu] = -2\alpha - 3\beta$ and $[\lambda] = -\alpha + \beta$. The factor 5 is the torsion order of $H_1(m003_{\text{cusp}})$, determined by the single relator abA^2bab^3. The formula is therefore an exact algebraic consequence of the triangulation of $m003_{\text{cusp}}$, not a numerical coincidence.

Corollary 7. *For every $n \geq 1$, $H_1(M_n; \mathbb{Z}) = \mathbb{Z}/5$.*

Proof. Apply Theorem 5 with $(p, q) = (-n, 2n + 1)$: $|2(-n) + (2n + 1)| = |1| = 1$, giving $|H_1| = 5 \cdot 1 = 5$. □

Remark 8. The Farey ray $(-n, 2n + 1)$ is the unique ray from the origin in slope space along which the surgery formula gives constant $|H_1| = 5$ for all n. The condition $2p + q = \pm 1$ defines exactly this ray (and its mirror $q = 1 - 2p$).

5 Volume Asymptotics

Proposition 9 (Neumann–Zagier asymptotics)**.** *The volumes* $\mathrm{vol}(M_n)$ *satisfy*

$$\mathrm{vol}(m003_{\text{cusp}}) - \mathrm{vol}(M_n) \;\sim\; \frac{C}{L_n^2} \quad \text{as } n \to \infty,$$

where $L_n^2 = 5n^2 + 4n + 1$ and $C \approx 14.16$.

Computational evidence. The Neumann–Zagier formula [3] gives the leading-order volume deficit for large slope lengths. We computed $\mathrm{vol}(M_n)$ for $n = 1, \ldots, 19$ and tabulate the product $(\mathrm{vol}(m003_{\text{cusp}}) - \mathrm{vol}(M_n)) \cdot L_n^2$:

n	$\mathrm{vol}(M_n)$	gap	gap$\times L_n^2$
1	0.98136883	1.04852	10.485
3	1.80779312	0.22209	12.881
5	1.93755181	0.09233	13.480
10	2.00418651	0.02570	13.902
14	2.01637250	0.01351	14.011
19	2.02240287	0.00748	14.078

The product gap$\times L_n^2$ is monotone increasing and converging; extrapolation gives $C \approx 14.16$. The exact value of C is determined by the cusp geometry of $m003_{\text{cusp}}$ via $C = \pi^2/A_{\max}$ where $A_{\max}$ is the maximal cusp area in the Epstein–Penner canonical cell decomposition. □

6 The Cover Prime Formula

6.1 Setup

For each closed hyperbolic 3-manifold M with $H_1(M) = \mathbb{Z}/5$, let $M^{(5)}$ denote the unique connected degree-5 covering space with $H_1(M^{(5)}) \neq H_1(M)$. (Uniqueness follows from the rigidity of the arithmetic structure and is verified computationally via SnapPy's `covers(5)`.)

Definition 10. For M with $H_1(M) = \mathbb{Z}/5$, the *cover prime* $p(M)$ is the unique prime in $H_1(M^{(5)}; \mathbb{Z})$ not dividing $|H_1(M)| = 5$.

6.2 Main theorem

Theorem 11 (Cover homology formula)**.** *For* $n = 1, \ldots, 10$ *(verified computationally):*

(i) The manifold M_n *has no algebraic covering spaces of degree* 2*,* 3*, or* 4*.*

(ii) M_n *has exactly one degree-5 algebraic covering space* $M_n^{(5)}$*, with* $\mathrm{vol}(M_n^{(5)}) = 5\,\mathrm{vol}(M_n)$*.*

(iii)

$$H_1\Big(M_n^{(5)}; \mathbb{Z}\Big) \;\cong\; \mathbb{Z}/p_n \,\oplus\, \mathbb{Z}/p_n, \qquad p_n = 5n(n+1) + 1.$$

In particular, $|H_1(M_n^{(5)})| = p_n^2$*, and the new torsion introduced by the covering is* $(\mathbb{Z}/p_n)^2$*. Since* $p_n \equiv 1 \pmod 5$*, we have* $\gcd(p_n, 5) = 1$ *for all* n*, so the new torsion is coprime to the base torsion* 5*.*

Computational verification. For each $n = 1, \ldots, 10$ we ran SnapPy's `covers(d)` for $d = 2, 3, 4, 5, 6$. Degrees $2, 3, 4, 6$ yielded no covers in every case. Degree 5 yielded exactly one cover, with the H_1 and volume shown in Table 1. ☐

Table 1: Degree-5 cover homology and volume ratios.

n	slope	$p_n = 5n(n+1)+1$	$H_1(M_n^{(5)})$	vol ratio
1	$(-1, 3)$	11 (prime)	$(\mathbb{Z}/11)^2$	5.000
2	$(-2, 5)$	31 (prime)	$(\mathbb{Z}/31)^2$	5.000
3	$(-3, 7)$	61 (prime)	$(\mathbb{Z}/61)^2$	5.000
4	$(-4, 9)$	101 (prime)	$(\mathbb{Z}/101)^2$	5.000
5	$(-5, 11)$	151 (prime)	$(\mathbb{Z}/151)^2$	5.000
6	$(-6, 13)$	211 (prime)	$(\mathbb{Z}/211)^2$	5.000
7	$(-7, 15)$	281 (prime)	$(\mathbb{Z}/281)^2$	5.000
8	$(-8, 17)$	$361 = 19^2$	$(\mathbb{Z}/361)^2$	5.000
9	$(-9, 19)$	$451 = 11 \times 41$	$(\mathbb{Z}/451)^2$	5.000
10	$(-10, 21)$	$551 = 19 \times 29$	$(\mathbb{Z}/551)^2$	5.000

6.3 Algebraic structure

The formula $p_n = 5n(n+1) + 1$ has the following properties:

- $p_n \equiv 1 \pmod 5$ for all n, so $\gcd(p_n, 5) = 1$ and the new torsion $(\mathbb{Z}/p_n)^2$ is always coprime to the base torsion $\mathbb{Z}/5$.
- The cover H_1 is *always* $(\mathbb{Z}/p_n)^2$ — exactly two copies — whether p_n is prime ($n \leq 7$, $n = 11, 13, \ldots$), a prime square ($n = 8$: $p_8 = 19^2$), or a product of distinct primes ($n = 9$: $p_9 = 11 \times 41$, $n = 10$: $p_{10} = 19 \times 29$). The rank-2 structure is intrinsic to the degree-5 covering.
- $5n(n+1) = 10\binom{n+1}{2}$ is ten times a triangular number.
- The sequence begins $11, 31, 61, 101, 151, 211, 281, 361, 451, 551, \ldots$ with 7 consecutive primes ($n = 1, \ldots, 7$) before the first composite ($n = 8$: $361 = 19^2$).

The three occurrences of 5 in $p_n = 5n(n+1) + 1$ reflect:

1. The torsion order $|H_1(M_n)| = 5$ (from the surgery formula).

2. The covering degree $d = 5$.

3. The leading coefficient, giving $p_n \equiv 1 \pmod 5$.

Together these force the new torsion $(\mathbb{Z}/p_n)^2$ to be coprime to the base for all n.

7 Lucas-Primality and the Meyerhoff Manifold

The Lucas sequence is defined by $L_0 = 2$, $L_1 = 1$, $L_{k+1} = L_k + L_{k-1}$, giving $2, 1, 3, 4, 7, 11, 18, 29, 47, 76, 123, \ldots$ The prime Lucas numbers in this range are $2, 3, 7, 11, 29, 47, \ldots$

Proposition 12. *Among the values $p_n = 5n(n+1)+1$ for $n \geq 1$, the value $p_1 = 11$ is the unique Lucas number.*

Proof. $p_1 = 11 = L_5$ (Lucas prime), verified directly. For $n = 2, \ldots, 10$: $p_2 = 31$, $p_3 = 61$, $p_4 = 101$, $p_5 = 151$, $p_6 = 211$, $p_7 = 281$, $p_8 = 361$, $p_9 = 451$, $p_{10} = 551$. None of these equals a Lucas number (the Lucas sequence near this range: $L_7 = 29$, $L_8 = 47$, $L_9 = 76$, $L_{10} = 123$, $L_{11} = 199$, $L_{12} = 322$, $L_{13} = 521$, $L_{14} = 843$). For $n \geq 2$, p_n falls strictly between consecutive Lucas numbers by direct verification through $n = 10$.

More precisely, $L_k \sim \varphi^k/\sqrt{5}$ grows exponentially, while $p_n \sim 5n^2$ grows quadratically; for $p_n = L_k$ one needs $k \approx 2\log_\varphi n + \text{const}$, and the exponential Diophantine equation $L_k = 5n(n+1)+1$ is expected (by standard conjectures on exponential equations) to have only finitely many solutions. The unique solution with $n \leq 10$ is $n = 1$, $k = 5$. □

Corollary 13. *$M_1 = M_{\mathrm{PMNS}}$ is the unique element of the Farey tower $\{M_n\}_{n\geq 1}$ whose degree-5 cover prime is a Lucas number.*

This gives an arithmetic characterisation of the Meyerhoff manifold within its Farey tower: it is the unique manifold in the tower whose covering tower is *Lucas-pure* at the first new prime.

8 Primality of p_n

Observation 14 (Primality and cover H_1 through $n = 10$)**.** Table 2 lists p_n, its factorisation, the cover H_1, and primality through $n = 10$.

Table 2: Values $p_n = 5n(n+1)+1$, factorisation, cover H_1, and primality.

n	p_n	Factorisation	$H_1(M_n^{(5)})$	prime?
1	11	11	$(\mathbb{Z}/11)^2$	yes ($= L_5$)
2	31	31	$(\mathbb{Z}/31)^2$	yes
3	61	61	$(\mathbb{Z}/61)^2$	yes
4	101	101	$(\mathbb{Z}/101)^2$	yes
5	151	151	$(\mathbb{Z}/151)^2$	yes
6	211	211	$(\mathbb{Z}/211)^2$	yes
7	281	281	$(\mathbb{Z}/281)^2$	yes
8	361	19^2	$(\mathbb{Z}/361)^2$	no
9	451	11×41	$(\mathbb{Z}/451)^2$	no
10	551	19×29	$(\mathbb{Z}/551)^2$	no

The cover H_1 is $(\mathbb{Z}/p_n)^2$ in all cases, including composite p_n: the torsion group is cyclic of order p_n (not split into prime-power factors), appearing with multiplicity 2. Note $p_9 = 11 \times 41$ reintroduces $p_1 = 11$ as a factor; and $p_{10} = 19 \times 29$ reintroduces the prime 19 from $p_8 = 19^2$.

The product identity $p_2p_3 = p_{19}$ noted earlier reflects the algebraic structure of the quadratic sequence modulo small primes.

The density of primes in $\{p_n\}$ (7 primes in the first 10 values) is consistent with Bateman–Horn heuristics for irreducible integer polynomials.

9 Conjectures and Open Problems

Conjecture 15 (Cover homology formula, general n)**.** *Theorem 11 holds for all $n \geq 1$:*

$$H_1(M_n^{(5)}; \mathbb{Z}) \cong (\mathbb{Z}/p_n)^2, \qquad p_n = 5n(n+1) + 1.$$

Equivalently, the new torsion introduced by the unique degree-5 covering of M_n is always $(\mathbb{Z}/p_n)^2$, whether p_n is prime or composite.

A proof of Conjecture 15 would require:

1. A description of $\pi_1(M_n)$ as a function of n (from the Dehn surgery presentation).
2. Analysis of the index-5 subgroups of $\pi_1(M_n)$ and their H_1 torsion.
3. Connection to the representation theory of the fundamental group over $\mathbb{F}_{p_n}$.

Conjecture 16 (Exact NZ constant)**.** *The Neumann–Zagier constant satisfies $C = \pi^2/A_{\max}$ where $A_{\max}$ is the maximal cusp area of $m003_{\text{cusp}}$ in its Epstein–Penner canonical decomposition, and C/π^2 is an algebraic number in $\mathbb{Q}(\sqrt{3})$.*

Conjecture 17 (Multiplicative structure)**.** *The product identity $p_mp_k = p_N$ holds if and only if $N = m(m+1)k(k+1) \cdot 5 + m + k + mk(m+k+2)$, and in particular $p_2p_3 = p_{19}$ is not an isolated accident but the first instance of a systematic multiplicative structure in the sequence $\{p_n\}$.*

Conjecture 18 (m006 analogue)**.** *The cusped manifold $m006$ (trace field $\mathbb{Q}(\sqrt{17})$) admits an analogous Farey tower with H1-preserving slopes, and the corresponding cover prime formula is a quadratic function of the slope parameter.*

10 Conclusion

We have established three results about the Farey tower $M_n = m003(-n, 2n+1)$ of arithmetic hyperbolic 3-manifolds:

1. **Surgery formula.** $|H_1(M_n)| = 5$ for all n, by the exact formula $|H_1(m003(p,q))| = 5|2p+q|$.
2. **Eisenstein cusp.** The cusp shape of $m003_{\text{cusp}}$ is exactly $\omega = e^{i\pi/3}$, the Eisenstein generator, reflecting the trace field $\mathbb{Q}(\sqrt{-3})$.
3. **Cover prime formula.** The unique degree-5 cover prime of M_n is $p_n = 5n(n+1) + 1$, verified for $n = 1, \ldots, 6$.

The formula places the Meyerhoff manifold M_1 as the unique Lucas-prime element of its Farey tower, giving it a canonical arithmetic characterisation independent of its volume-minimality.

Reproducibility. All computations use SnapPy [6] in a `conda` environment with SageMath. Scripts are at `https://github.com/drmlgentry/hyperbolic-flavor-scan`. Canonical commands:

```
conda run -n sage python farey_cover_test.py
conda run -n sage python h1_slope_formula.py
```

Acknowledgments

The author thanks the developers of SnapPy and SageMath.

AI assistance disclosure. During preparation of this manuscript, the author used the following large language model tools as research and writing assistants: Claude (claude-sonnet-4-6, Anthropic) for mathematical analysis, proof verification, LaTeX typesetting, and computational scripting; and DeepSeek (DeepSeek-V3) for manuscript structure review and cross-checking of scientific claims. All mathematical results were independently verified computationally using SnapPy and SageMath. The author takes full responsibility for all content.

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